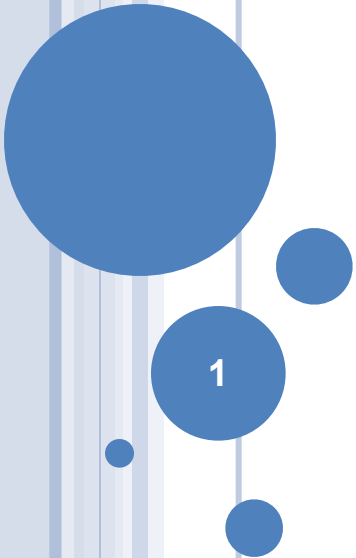


INTRODUCTION TO PROBABILITY AND STATISTICS FOURTEENTH EDITION



Chapter 12 Linear Regression and Correlation

1

CH11 VS CH12?



- In Chapter 11, we used ANOVA to investigate the effect of various **factor-level combinations** (treatments) on a **response**.
 - Our objective was to see whether the treatment means were different.
- In Chapters 12 and 13, we investigate a response y which is affected by various **independent variables**, x_i .

INTRODUCTION

- Our objective is to use the information provided by the x_i to predict the value of y .
- We plot the value of x in the x -axis, and the value of y in the y -axis.
 - x , factor, explanatory variable, independent variable, input
 - y , response variable, dependent variable, output

SUM OF SQUARES: DEFINITIONS AND SIMPLIFICATIONS

$$S_{xx} := \sum (x_i - \bar{x})^2 = \sum (x_i^2 - 2\bar{x}x_i + \bar{x}^2) = \sum x_i^2 - 2\bar{x}(n\bar{x}) + n\bar{x}^2 = \sum x_i^2 - n\bar{x}^2$$

$$S_{yy} := \sum (y_i - \bar{y})^2 = \sum (y_i^2 - 2\bar{y}y_i + \bar{y}^2) = \sum y_i^2 - 2\bar{y}(n\bar{y}) + n\bar{y}^2 = \sum y_i^2 - n\bar{y}^2$$

$$S_{xy} := \sum (x_i - \bar{x})(y_i - \bar{y}) = \sum (x_i y_i - \bar{x}y_i - \bar{y}x_i + \bar{x}\bar{y}) = \sum x_i y_i - \bar{x}(n\bar{y}) - \bar{y}n\bar{x} + n\bar{x}\bar{y} = \sum x_i y_i - n\bar{x}\bar{y}.$$

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12.1 SIMPLE LINEAR REGRESSION

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EXAMPLE



- Let y be a student's college achievement, measured by his/her **GPA**. This might be a function of several variables:
 - x_1 = rank in high school class
 - x_2 = high school's overall rating
 - x_3 = high school GPA
 - x_4 = SAT scores
- We want to predict y using knowledge of x_1 , x_2 , x_3 and x_4 .

EXAMPLE



- Let y be the **monthly sales revenue** for a company. This might be a function of several variables:
 - x_1 = advertising expenditure
 - x_2 = time of year
 - x_3 = state of economy
 - x_4 = size of inventory
- We want to predict y using knowledge of x_1 , x_2 , x_3 and x_4 .

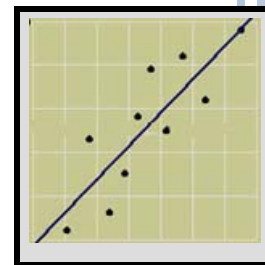
SOME QUESTIONS



- Which of the independent variables are useful and which are not?
- How could we create a prediction equation to allow us to predict y using knowledge of x_1, x_2, x_3 etc?
- How good is this prediction?

We start with the simplest case, in which the response y is a function of a single independent variable, x .

A SIMPLE LINEAR MODEL



- In Chapter 3, we used the equation of a line to describe the relationship between y and x for a **sample** of n pairs, (x, y) .
- If we want to describe the relationship between y and x for the **whole population**, there are two models we can choose

- **Deterministic (Economic) Model:** $y = \alpha + \beta x$

- **Probabilistic (Econometric) Model:**

- $y = \text{deterministic model} + \text{random error}$

- $y = \alpha + \beta x + \varepsilon$

A SIMPLE LINEAR MODEL

- Since the bivariate measurements that we observe do not generally fall **exactly** on a straight line, we choose to use:

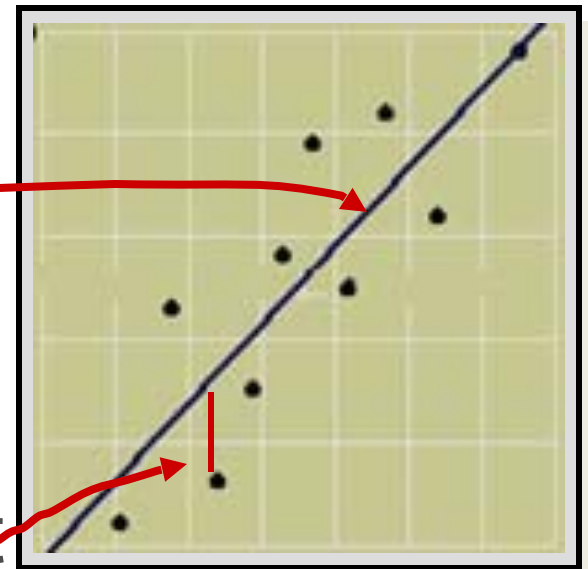
- **Probabilistic Model:**

- $y = \alpha + \beta x + \varepsilon$

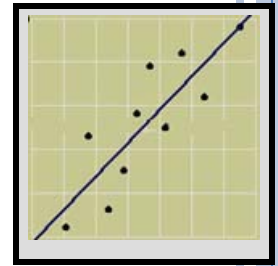
- $E(y | x) = \alpha + \beta x$

Points deviate from the **line of means** by an amount

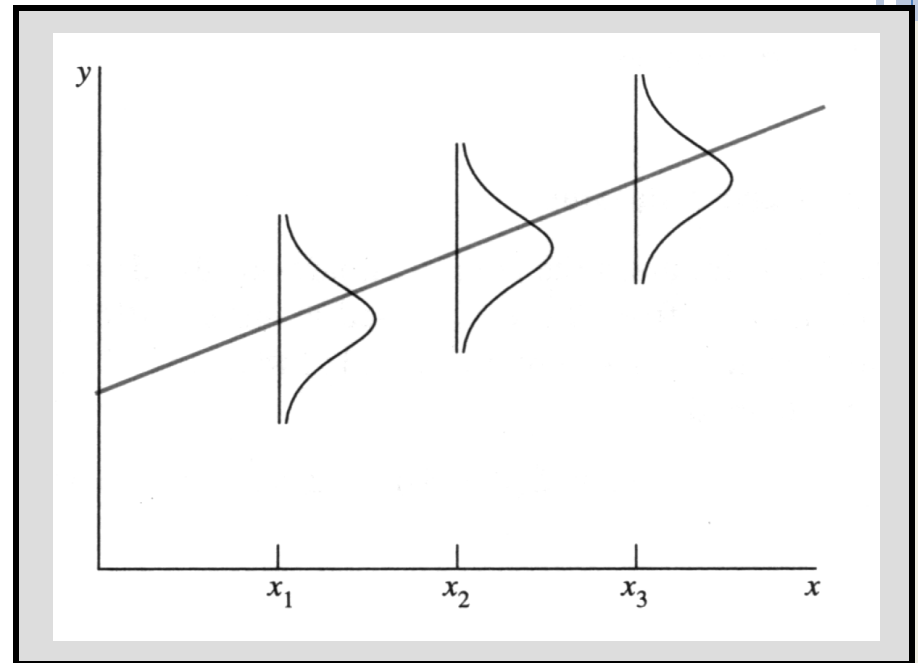
ε where ε has a normal distribution with mean 0 and variance σ^2 .



THE RANDOM ERROR



- The line of means, $E(y | x) = \alpha + \beta x$, describes average value of y for any fixed value of x .
- The population of measurements is generated as y deviates from the population line by ε . We estimate α and β using sample information.



$$E(y) := E(y | x) = \alpha + \beta x$$

The line of means

$$E(y) := E(y | x) = \alpha + \beta x,$$

is also called:

- the conditional mean given x :
- the conditional expectation given x ,
- the simple regression function
- the regression function

$$E(y) := E(y | x) = \alpha + \beta x$$

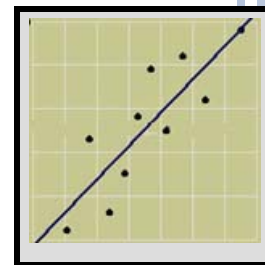
The line of means

$$E(y) := E(y | x) = \alpha + \beta x,$$

Interpretations of α and β

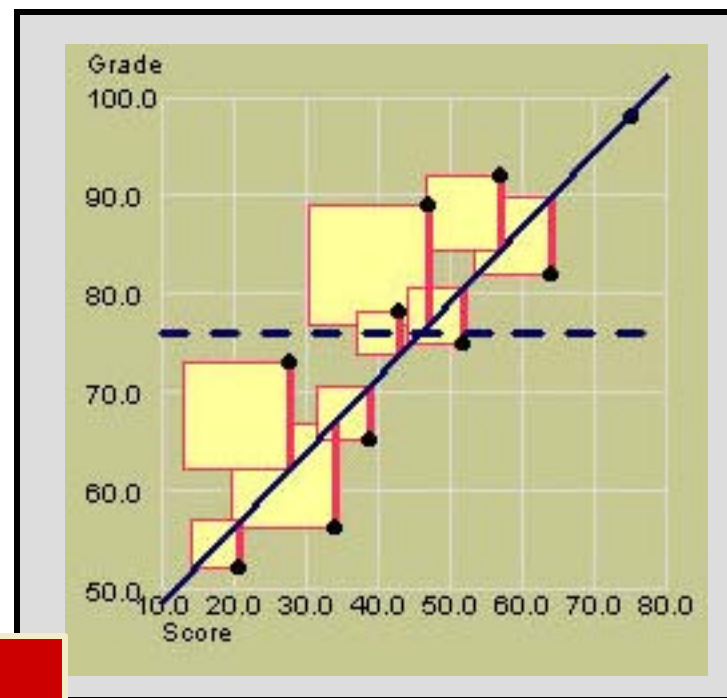
- α : when $x = 0$, on average, y is α .
- 當 $x=0$, y 平均而言是 α
- β : when x increases one unit, on average, y increases β unit.
- 當 x 增加一個單位，平均而言， y 增加 β 個單位。

THE METHOD OF LEAST SQUARES



○ The equation of the best-fitting line is calculated using a set of n pairs (x_i, y_i) .

• We choose our estimates a and b to estimate α and β so that the vertical distances of the points from the line, are minimized.



Bestfitting line: $\hat{y} = a + bx$

Choose a and b to minimize

$$SSE = \sum (y - \hat{y})^2 = \sum (y - a - bx)^2$$

APPENDIX: PROOF OF LSE

- Least squares estimators a and b minimize $\text{SSE}(a, b)$, where

$$\text{SSE}(a, b) = \sum (y - \hat{y})^2 = \sum_{i=1}^n (y_i - a - bx_i)^2.$$

- Results: The least squares estimators are:

$$b = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{S_{xy}}{S_{xx}},$$

$$a = \bar{y} - b\bar{x}.$$

- The *estimated* or *fitted* regression line is:

$$\hat{y}_i = a + bx_i.$$



EXAMPLE

The table shows the math achievement test scores for a random sample of $n = 10$ college freshmen, along with their final calculus grades.

Student	1	2	3	4	5	6	7	8	9	10
Math test, x	39	43	21	64	57	47	28	75	34	52
Calculus grade, y	65	78	52	82	92	89	73	98	56	75

Use your calculator to find the sums and sums of squares.

$$\begin{aligned}\sum x &= 460 & \sum y &= 760 \\ \sum x^2 &= 23634 & \sum y^2 &= 59816 \\ \sum xy &= 36854 \\ \bar{x} &= 46 & \bar{y} &= 76\end{aligned}$$

EXAMPLE

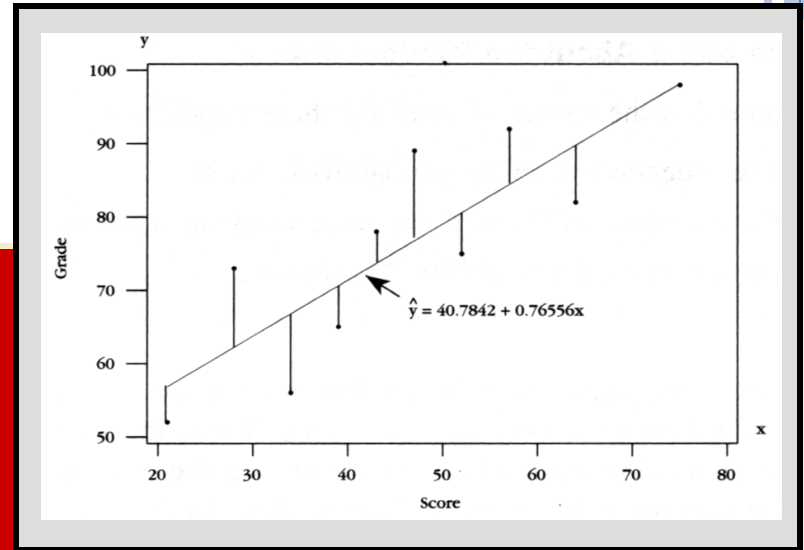
$$S_{xx} = 23634 - \frac{(460)^2}{10} = 2474$$

$$S_{yy} = 59816 - \frac{(760)^2}{10} = 2056$$

$$S_{xy} = 36854 - \frac{(460)(760)}{10} = 1894$$

$$b = \frac{1894}{2474} = .76556 \quad \text{and} \quad a = 76 - .76556(46) = 40.78$$

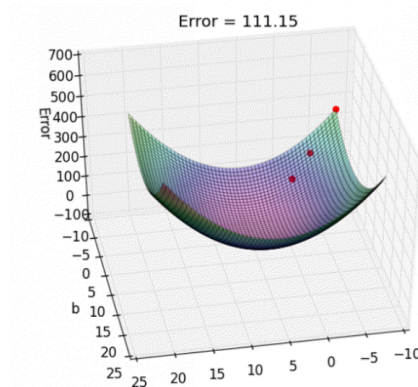
Bestfitting line: $\hat{y} = 40.78 + .77x$



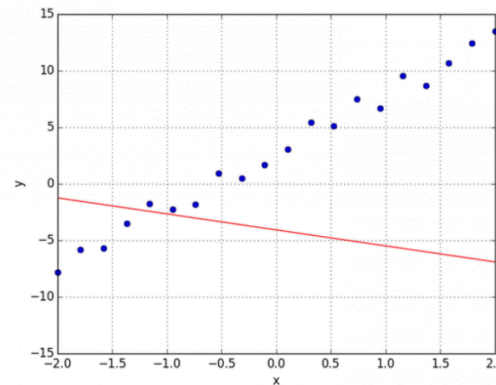
Demo.

$$\text{SSE}(a, b) = \sum_{i=1}^n (y_i - a - bx_i)^2.$$

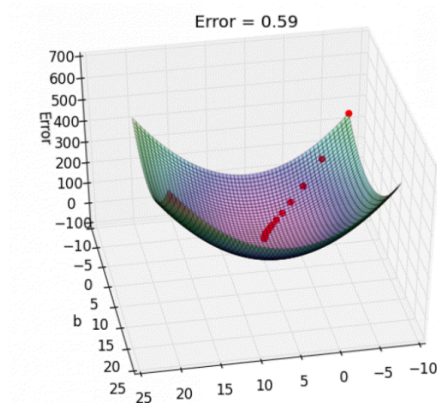
$$a = -1.41, b = -4.10$$



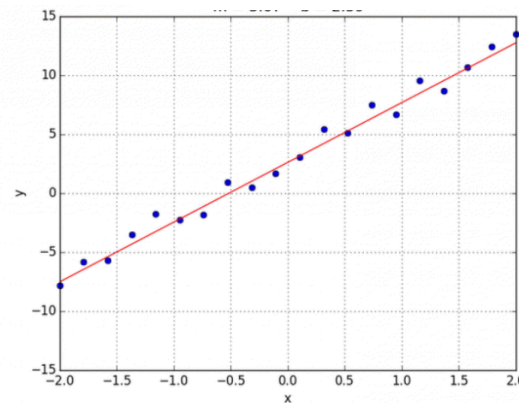
a



$$a = 5.07, b = 2.59$$



a



APPENDIX A: LSE formulas derivations

- Recall that $SSE(a, b) = \sum_{i=1}^n (y_i - a - bx_i)^2$. First-order-Condition:

$$\frac{\partial SSE(a, b)}{\partial a} = -2 \sum (y_i - a - bx_i) = 0,$$
$$\frac{\partial SSE(a, b)}{\partial b} = -2 \sum x_i(y_i - a - bx_i) = 0.$$

- Simple math gives

$$\sum y_i = na + (\sum x_i)b, \quad (1)$$

$$\sum x_i y_i = (\sum x_i)a + (\sum x_i^2)b. \quad (2)$$

- Multiplying (1) by $(\sum x_i)$ and multiply (2) by n , we have

$$(\sum x_i)(\sum y_i) = n(\sum x_i)a + (\sum x_i^2)b, \quad (3)$$

$$n(\sum x_i y_i) = n(\sum x_i)a + n(\sum x_i^2)b. \quad (4)$$

- Thus, (4)-(3) gives

$$b = \frac{n(\sum x_i y_i) - (\sum x_i \sum y_i)}{n(\sum x_i^2) - (\sum x_i)^2} = \frac{\sum x_i y_i - n\bar{x}\bar{y}}{\sum x_i^2 - n\bar{x}^2} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{S_{xy}}{S_{xx}}.$$

- Plug b back to (1), we have $a = \bar{y} - \bar{x}b$.

The left side of the slide features a series of vertical stripes in various shades of blue and white. Overlaid on these stripes are several blue circles of different sizes. One large circle is positioned near the top left, and several smaller circles are arranged vertically below it, some overlapping the stripes.

12.2 AN ANALYSIS OF VARIANCE FOR LIENAR REGRESSION

20



THE ANOVA TABLE

Total $df = n - 1$

Regression $df = 1$

Error $df = n - 1 - 1 = n - 2$

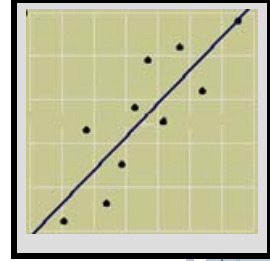
Mean Squares

$MSR = SSR / (1)$

$MSE = SSE / (n - 2)$

Source	df	SS	MS	F
Regression	1	SSR	$MSR = SSR / (1)$	$F = MSR / MSE$
Error	$n - 2$	SSE	$MSE = SSE / (n - 2)$	
Total	$n - 1$	Total SS		

THE ANALYSIS OF VARIANCE



The total variation in the experiment is measured by the **total sum of squares**:

$$\text{Total SS} = S_{yy} = \sum (y - \bar{y})^2$$

The **Total SS** is divided into two parts:

✓ **SSR** (sum of squares for regression):
measures the variation explained by using x in the model.

✓ **SSE** (sum of squares for error):
measures the leftover variation not explained by x .

TOTAL SS

- Total SS = $\sum (y_i - \bar{y})^2 = S_{yy}$. Note that $\sum_{i=1}^n y_i = n\bar{y}$. Thus,

$$\begin{aligned}\sum_{i=1}^n (y_i - \bar{y})^2 &= \sum_{i=1}^n (y_i^2 - 2\bar{y}y_i + \bar{y}^2) = \sum_{i=1}^n y_i^2 - 2\bar{y} \sum_{i=1}^n y_i + n\bar{y}^2 \\ &= \sum_{i=1}^n y_i^2 - 2\bar{y}(n\bar{y}) + n\bar{y}^2 = \sum_{i=1}^n y_i^2 - n\bar{y}^2\end{aligned}$$

HOW TO CALCULATE ANOVA TABLE?

1. Total SS = $\sum (y - \bar{y})^2 = S_{yy}$.

2. SSR

$$\begin{aligned}\text{SSR} &:= \sum (\hat{y}_i - \bar{y})^2 = \sum ((a + bx_i) - \bar{y})^2 = \sum ((\bar{y} - b\bar{x}) + bx_i - \bar{y})^2 \\ &= \sum (b(x_i - \bar{x}))^2 = b^2 S_{xx} = \left(\frac{S_{xy}}{S_{xx}} \right)^2 S_{xx} = \frac{S_{xy}^2}{S_{xx}}.\end{aligned}$$

3. SSE = Total SS – SSR

THE ANALYSIS OF VARIANCE

We calculate

$$SSR = \frac{(S_{xy})^2}{S_{xx}} = \frac{1894^2}{2474}$$

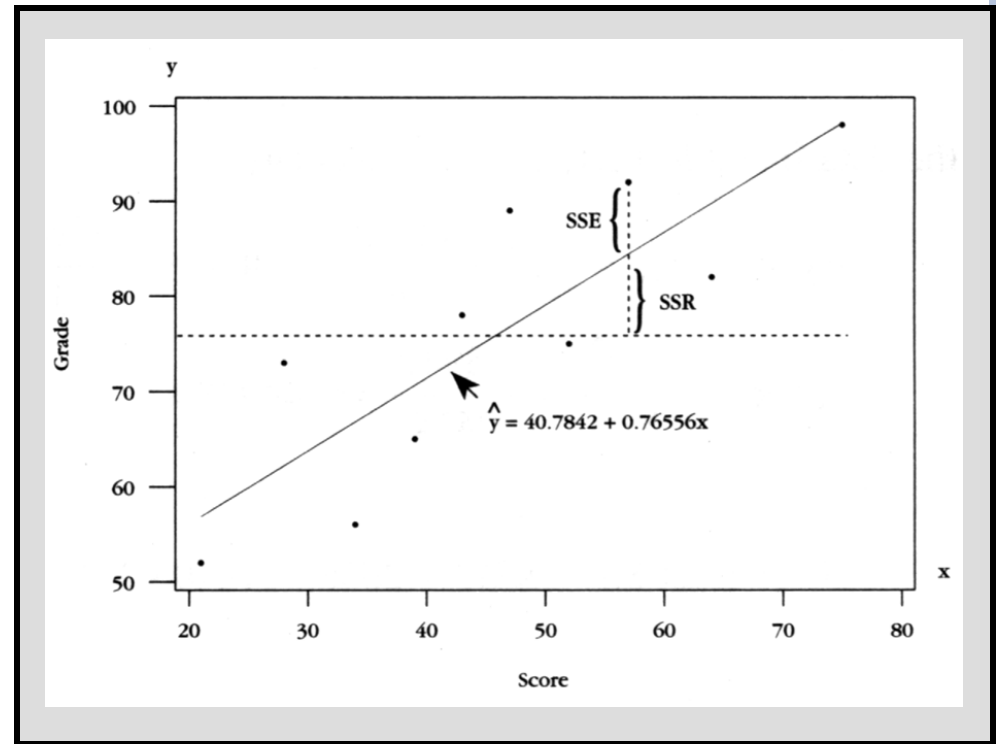
$$= 1449.9741$$

$$SSE = \text{Total SS} - SSR$$

$$= S_{yy} - \frac{(S_{xy})^2}{S_{xx}}$$

$$= 2056 - 1449.9741$$

$$= 606.0259$$



THE CALCULUS PROBLEM



$$SSR = \frac{(S_{xy})^2}{S_{xx}} = \frac{1894^2}{2474} = 1449.9741$$

$$SSE = \text{Total SS} - SSR = S_{yy} - \frac{(S_{xy})^2}{S_{xx}} \\ = 2056 - 1449.9741 = 606.0259$$

Source	df	SS	MS	F
Regression	1	1449.9741	1449.9741	19.14
Error	8	606.0259	75.7532	
Total	9	2056.0000		

CALCULATION OF CH 12.1-12.2

- A: Calculate $(\sum x_i)$, $(\sum y_i)$, $(\sum x_i^2)$, $(\sum y_i^2)$, $(\sum x_i y_i)$, $\bar{x}$, and $\bar{y}$
- B: Calculate
 - $S_{xx} := \sum (x - \bar{x})^2 = \sum x^2 - (\sum x)^2/n$.
 - $S_{yy} := \sum (y - \bar{y})^2 = \sum y^2 - (\sum y)^2/n$.
 - $S_{xy} := \sum (x - \bar{x})(y - \bar{y}) = \sum xy - (\sum x)(\sum y)/n$.
- C: Calculate
 - $b = S_{xy}/S_{xx}$
 - $a = \bar{y} - b\bar{x}$
 - Best fitting line: $y = a + bx$
- D: Complete the ANOVA table:
 - (1) totalSS = S_{yy}
 - (2) SSR = S_{xy}^2/S_{xx} ,
 - (3) SSE = totalSS – SSR.

Next, Hypothesis Testing!

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12.3 TESTING THE USEFULNESS OF THE LINEAR REGRESSION MODEL

THREE EQUIVELENT APPROACHES

1. H_0 : model is not useful in explaining the response variable vs H_1 : model is useful in explaining the response variable

- Using ANOVA, F test, right-tailed test

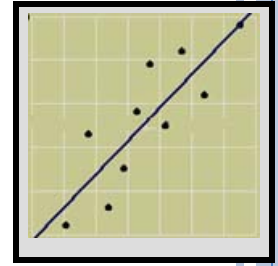
2. $H_0 : \beta = 0$ vs $H_1 : \beta \neq 0$

- $t_{STAT} = \frac{b}{\sqrt{MSE/S_{xx}}} \sim t_{df=(n-2)}$, t test, two-sided test

3. $H_0 : \rho = 0$ vs $H_1 : \rho \neq 0$

- $t_{STAT} = r \sqrt{\frac{n-2}{(1-r^2)}} \sim t_{(n-2)}$, t test, two-sided test

TESTING THE USEFULNESS OF THE MODEL



- The first question to ask is whether the independent variable x is of any use in predicting y .
- If it is not, then the value of y does not change, regardless of the value of x . This implies that the slope of the line, β , is zero.

$$H_0 : \beta = 0 \quad \text{versus} \quad H_a : \beta \neq 0$$

KEY INGREDIENTS

$$a \sim N\left(\alpha, \frac{\sigma^2 \sum x_i^2}{n \sum (x_i - \bar{x})^2}\right),$$

- $b \sim N\left(\beta, \frac{\sigma^2}{\sum (x_i - \bar{x})^2}\right).$

- See appendix for derivations in p.44 - p.49.
- With the above information, we can do statistical inference:
 - Confidence interval
 - Hypothesis test

t -TEST

1. $H_0: \beta = 0$ vs $H_1: \beta \neq 0$
2. Set α
3. $t_{STAT} = \frac{b - 0}{\sqrt{MSE/S_{xx}}} \sim t_{(n-2)}$.
4. Calculate t^*
5. Find the rejection region or p -value. (two-tail test).
6. Conclude.

SUMMARY

- The $100(1 - \alpha)\%$ CI for β is

$$b \pm t_{(n-2);\alpha/2} \sqrt{\frac{\text{MSE}}{S_{xx}}}.$$

- To ask whether x is useful in predicting y , we set $H_0 : \beta = 0$ vs $H_1 : \beta \neq 0$. The test statistic is

$$t_{STAT} = \frac{b - 0}{\sqrt{\frac{\text{MSE}}{S_{xx}}}} \sim t_{(n-2)}.$$

THE CALCULUS PROBLEM



- Is there a significant relationship between the calculus grades and the test scores at the 5% level of significance?



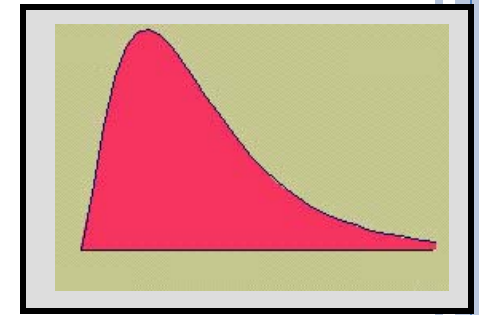
$$H_0 : \beta = 0 \text{ versus } H_a : \beta \neq 0$$

$$t = \frac{b - 0}{\sqrt{\text{MSE} / S_{xx}}} = \frac{.7656 - 0}{\sqrt{75.7532 / 2474}} = 4.38$$

Reject H_0 when $|t| > 2.306$. Since $t = 4.38$ falls into the rejection region, H_0 is rejected .

There is a significant linear relationship between the calculus grades and the test scores for the population of college freshmen.

THE F TEST



- You can test the overall usefulness of the model using an F test. If the model is useful, MSR will be large compared to the unexplained variation, MSE.

To test H_0 : model is useful in predicting y

$$\text{Test Statistic: } F = \frac{\text{MSR}}{\text{MSE}}$$

Reject H_0 if $F > F_\alpha$ with 1 and $n - 2$ *df*.

THE F TEST

You can test the overall usefulness of the model using an F test. If the model is useful, MSR will be large compared to the unexplained variation MSE.

1. H_0 : The model is not useful in predicting y vs H_1 :
The model is useful in predicting y

2. Set up α

3. Test statistic: $F_{STAT} = \frac{MSR}{MSE} \sim F_{1,(n-2)}$.

4. Calculate realized statistic F^* .

5. Find rejection region of p -value.

This is a right-tailed test!

6. Conclude

This test is exactly equivalent to the t -test, with $t^2 = F$.

MINITAB OUTPUT

Least squares
regression line

To test $H_0 : \beta = 0$



Regression Analysis: y versus x

The regression equation is $y = 40.8 + 0.766 x$

Predictor	Coef	SE Coef	T	P
Constant	40.784	8.507	4.79	0.001
x	0.7656	0.1750	4.38	0.002

$S = 8.70363$ $R\text{-Sq} = 70.5\%$ $R\text{-Sq}(\text{adj}) = 66.8\%$

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	1	1450.0	1450.0	19.14	0.002
Residual Error	8	606.0	75.8		
Total	9	2056.0			

$\sqrt{MSE}$

Regression
coefficients, a and b $t^2 = F$

WHY $t_{STAT}^2 = F_{STAT}$?

- The t test and F test are indeed the same! This is because:

$$\begin{aligned} t_{STAT}^2 &= \left(\frac{b}{\sqrt{MSE/S_{xx}}} \right)^2 = \frac{b^2}{MSE/S_{xx}} = \frac{b^2 S_{xx}}{MSE} \\ &= \frac{S_{xy}^2}{S_{xx}^2} \frac{S_{xx}}{MSE} \\ &= \frac{S_{xy}^2}{S_{xx} MSE} = \frac{SSR}{MSE} = \frac{SSR/1}{MSE} = \frac{MSR}{MSE} = F_{STAT} \end{aligned}$$

MEASURING THE STRENGTH OF THE RELATIONSHIP

- If the independent variable x is useful in predicting y , you will want to know how well the model fits.
- The strength of the relationship between x and y can be measured using:
 - Correlation coefficient: $r = \frac{s_{xy}}{\sqrt{s_{xx}s_{yy}}}$.
 - Coefficient of determination: $R^2 = \frac{\text{SSR}}{\text{Total SS}}$
(pronounced "R squared").

R^2

- Because $\text{Total} = \text{SS} + \text{SSE}$, R^2 measures the proportion of the total variation in the response that can be explained by using the explanatory variable x in the model.
- The percent reduction the total variation by using the regression equation rather than just using the sample mean $\bar{y}$ to estimate y .
- For the calculation problem, $R^2 = 0.705$ or 70.5%. The model is working well!

WHY $R^2 = r^2$?

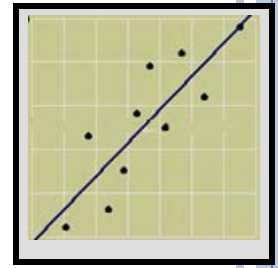
- This is because

$$r^2 = \left(\frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} \right)^2 = \frac{S_{xy}^2}{S_{xx}S_{yy}} = \frac{S_{xy}^2}{S_{xx}} \frac{1}{S_{yy}} = \frac{\text{SSR}}{\text{Total SS}} = R^2$$

-

Recall that $\text{SSR} = \frac{S_{xy}^2}{S_{xx}}!$

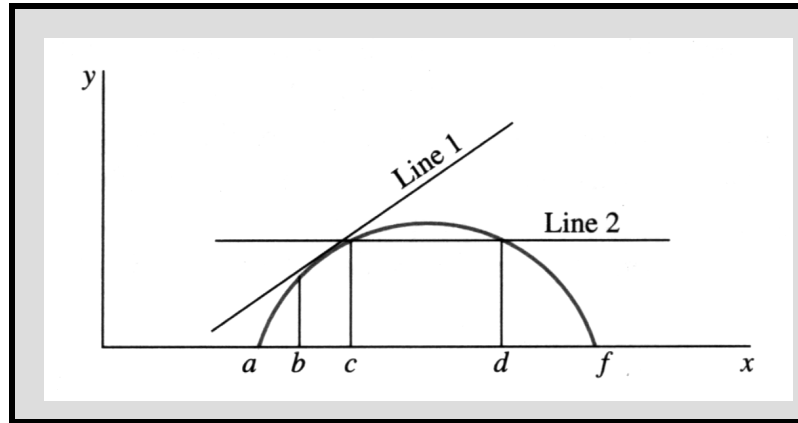
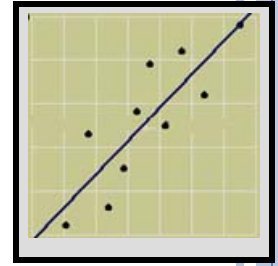
INTERPRETING A SIGNIFICANT REGRESSION



- Even if you do not reject the null hypothesis that the slope of the line equals 0, it does not necessarily mean that y and x are unrelated.
- **Type II error**—falsely declaring that the slope is 0 and that x and y are unrelated.
- It may happen that y and x are perfectly related in a **nonlinear** way.

SOME CAUTIONS

- You may have fit the wrong model.



- Extrapolation**—predicting values of y outside the range of the fitted data.
- Causality**—Do not conclude that x causes y . There may be an unknown variable at work!

APPENDIX B: A USEFUL FORMULA

- Linear combinations of independent normal distributions remain a normal distribution. Specifically, if $X_i \sim N(\mu_i, \sigma^2)$ and X_i are independent, then
- $$\sum a_i X_i \sim N\left(\sum a_i \mu_i, \sigma^2 \left(\sum a_i^2\right)\right).$$

APPENDIX: WHY $b \sim N(\beta, \frac{\sigma^2}{\sum (x_i - \bar{x})^2})$?

- Because $\sum (x_i - \bar{x}) = 0$, we rewrite b as

$$b = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$= \frac{\sum (x_i - \bar{x})y_i}{\sum (x_i - \bar{x})^2}$$

- $$= \sum \left(\frac{x_i - \bar{x}}{\sum (x_i - \bar{x})^2} \right) y_i$$

$$= \sum w_i y_i,$$

- where $w_i = \frac{x_i - \bar{x}}{\sum (x_i - \bar{x})^2}$.

- b is a linear combination of normal distribution, and hence a normal distribution.

- To find the expectation and variance of b , note the following identities:

- $\sum w_i = 0;$

- $\sum w_i x_i = \sum \frac{(x_i - \bar{x})x_i}{\sum (x_i - \bar{x})^2} = \sum \frac{(x_i - \bar{x})x_i - (x_i - \bar{x})\bar{x}}{\sum (x_i - \bar{x})^2} = \sum \frac{(x_i - \bar{x})(x_i - \bar{x})}{\sum (x_i - \bar{x})^2} = 1;$

- $\sum w_i^2 = \sum \left(\frac{(x_i - \bar{x})}{\sum (x_i - \bar{x})^2} \right)^2 = \frac{1}{\sum (x_i - \bar{x})^2}.$

- Rewrite

$$b = \sum w_i(\alpha + \beta x_i + e_i)$$

$$= \alpha \sum w_i + \beta \sum w_i x_i + \sum w_i e_i$$

- $$= \beta + \sum w_i e_i.$$

- Therefore,

$$E(b) = \beta$$

- $$\text{var}(b) = \sum w_i^2 \sigma^2 = \sigma^2 \frac{1}{\sum (x_i - \bar{x})^2}.$$

-

APPENDIX: WHY $a \sim N(\alpha, \sigma^2 \frac{\sum x_i^2}{n \sum (x_i - \bar{x})^2})$?

- Now, we write

$$a = \bar{y} - b\bar{x}$$

$$= \sum \left(\frac{1}{n} - \bar{x}w_i \right) y_i$$

$$= \sum \left(\frac{1}{n} - \bar{x}w_i \right) (\alpha + \beta x_i + e_i)$$

- $$= (\alpha - \alpha \bar{x} \sum w_i) + \left(\beta \bar{x} - \bar{x} \beta \sum x_i w_i \right) + \sum \left(\frac{1}{n} - \bar{x}w_i \right) e_i$$

$$= \alpha + \sum \left(\frac{1}{n} - \bar{x}w_i \right) e_i.$$

Hence, a is a normal distribution.

- To find the expectation and variance of a , it is easy to see

$$E(a) = \alpha + \sum \left[\left(\frac{1}{n} - \bar{x} w_i \right) 0 \right] = \alpha$$

$$\text{var}(a) = \sum \left(\frac{1}{n} - \bar{x} w_i \right)^2 \sigma^2$$

$$= \sigma^2 \left(\sum \frac{1}{n^2} - 2 \sum \frac{\bar{x}}{n} w_i + \bar{x}^2 \sum w_i^2 \right)$$

$$= \sigma^2 \left(\frac{1}{n} + \bar{x}^2 \frac{1}{\sum (x_i - \bar{x})^2} \right)$$

$$= \sigma^2 \frac{\sum (x_i - \bar{x})^2 + n \bar{x}^2}{n \sum (x_i - \bar{x})^2}$$

$$= \sigma^2 \frac{(\sum x_i^2 - 2n\bar{x}^2 + n\bar{x}^2) + n\bar{x}^2}{n \sum (x_i - \bar{x})^2} = \sigma^2 \frac{\sum x_i^2}{n \sum (x_i - \bar{x})^2}.$$

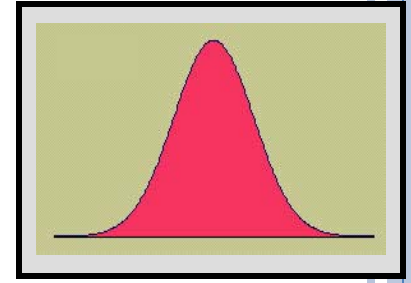
A decorative graphic on the left side of the slide. It features a vertical stack of stripes in various shades of blue and white. Overlaid on these stripes are several circles of different sizes, also in shades of blue. The largest circle is positioned near the top left, with several smaller circles arranged vertically below it and to its right.

12.4

50

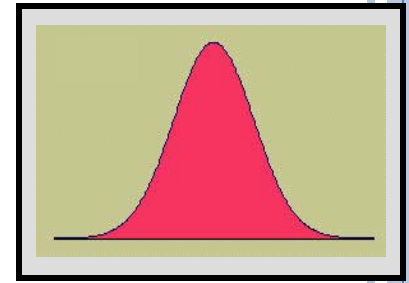
Diagnostic Tools for checking the regression assumptions (Residual analysis)

CHECKING THE REGRESSION ASSUMPTIONS



1. The relationship between x and y is linear, given by $y = \alpha + \beta x + \varepsilon$. ε : error 誤差
 2. The random error terms ε are independent and, for any value of x , have a normal distribution with mean 0 and variance σ^2 .
- Remember that the results of a regression analysis are only valid when the necessary assumptions have been satisfied.

DIAGNOSTIC TOOLS



- We use the same diagnostic tools used in Chapter 11 to check the normality assumption and the assumption of equal variances.

1. **Normal probability plot** of residuals
2. Plot of **residuals versus fit** or **residuals versus variables**

RESIDUALS

- Residual = 殘差, estimated error:

$$r_i := y_i - \hat{y}_i = y_i - a - bx_i.$$

- If all assumptions have been met, these residuals should be $N(0, \sigma^2)$.

- Informal way

1. Residuals plots (identical and independent)

- r_i vs $\hat{y}_i$

- r_i vs i

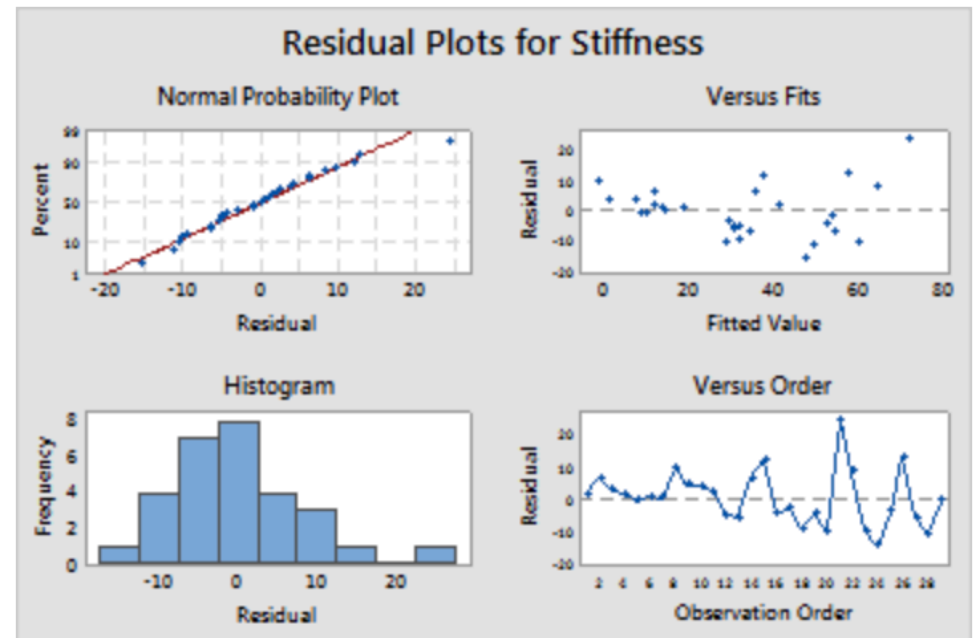
- r_i vs x_i

- Normal distribution

- Histogram

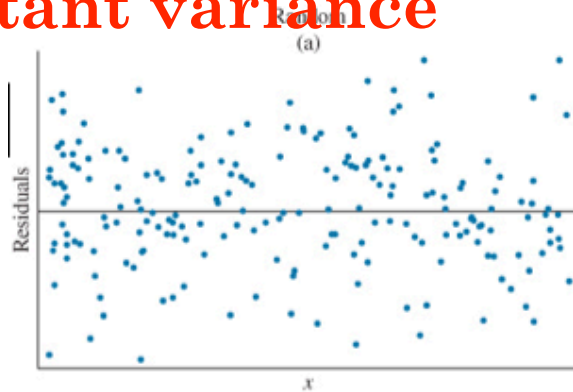
- Normal probability plot

The **residual error** is the “leftover” variation in each data point after the variation explained by the regression model has been removed.

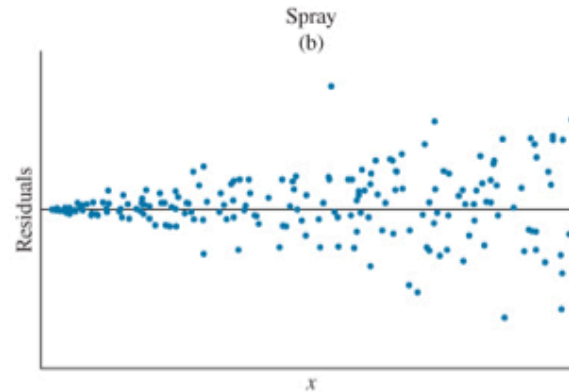


MORE EXAMPLES ON RESIDUAL PLOTS

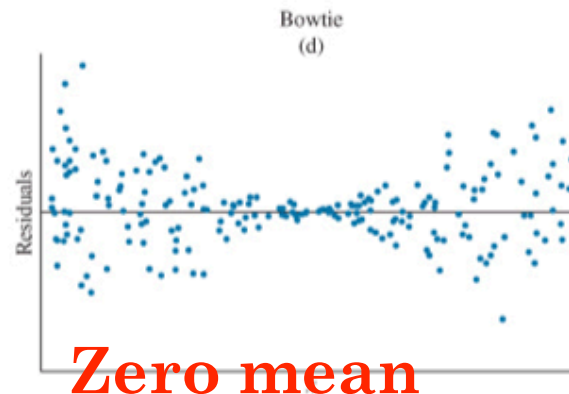
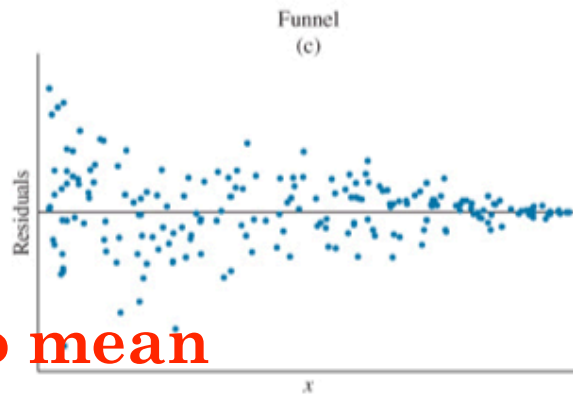
**Zero mean
constant variance**



**Zero mean
variance not constant**



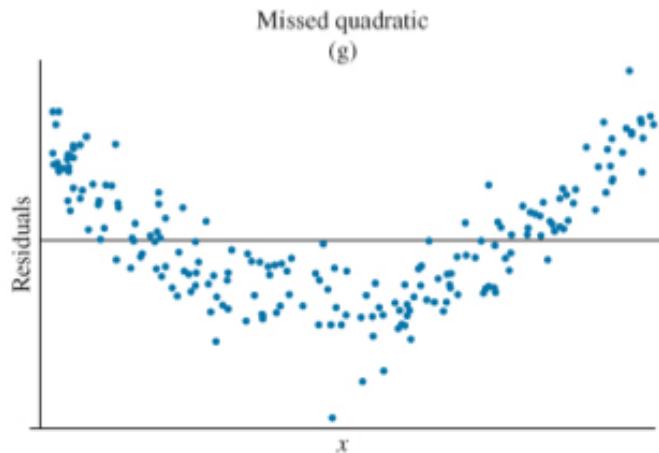
**Zero mean
variance not constant**



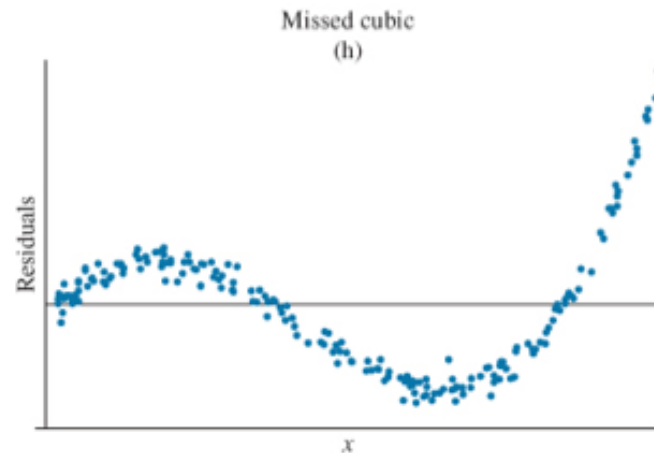
**Zero mean
variance not constant**

MORE EXAMPLES ON RESIDUAL PLOTS

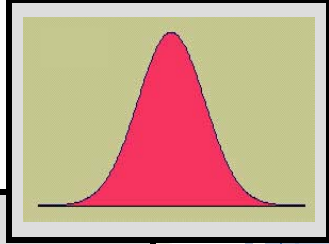
mean not constant zero



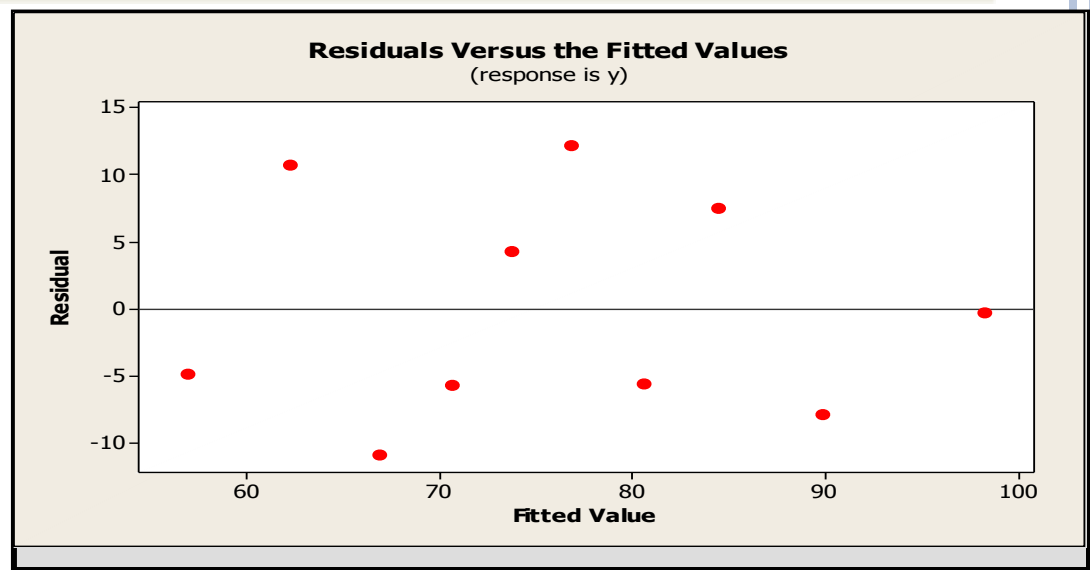
mean not constant zero



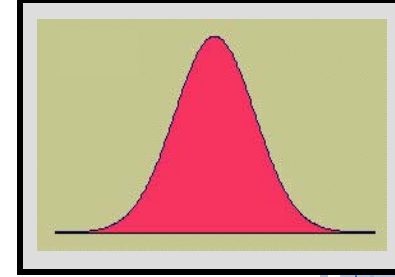
RESIDUALS VERSUS FITS



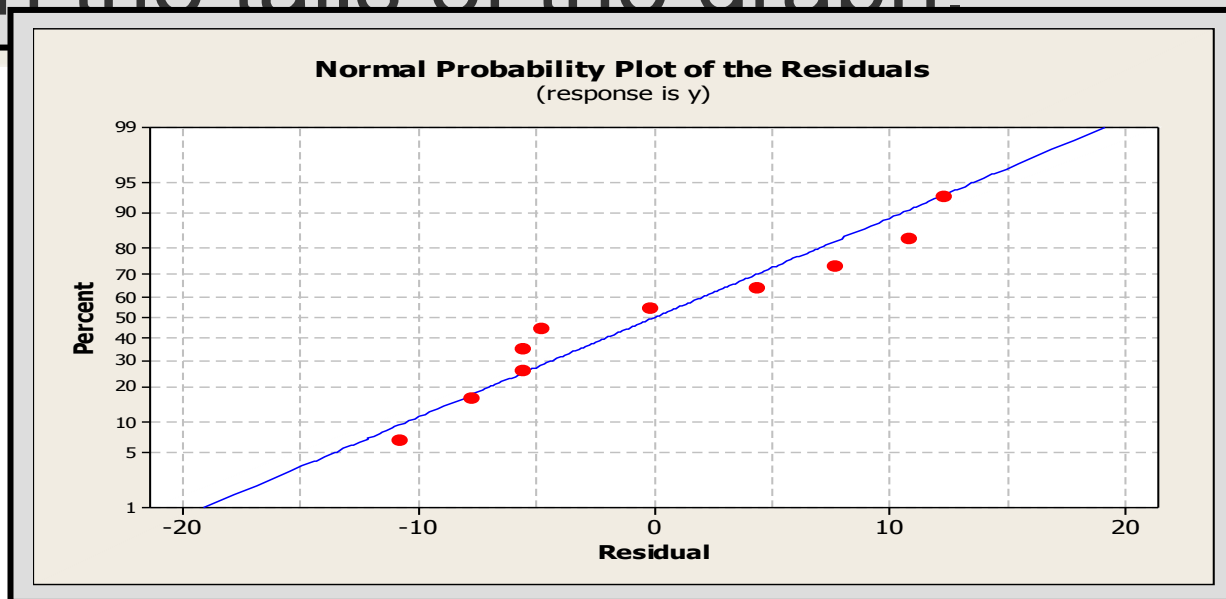
- ✓ If the equal variance assumption is valid, the plot should appear as a random scatter around the zero center line.
- ✓ If not, you will see a pattern in the residuals.



NORMAL PROBABILITY PLOT



- ✓ If the normality assumption is valid, the plot should resemble a straight line, sloping upward to the right.
- ✓ If not, you will often see the pattern fail in the tails of the graph.

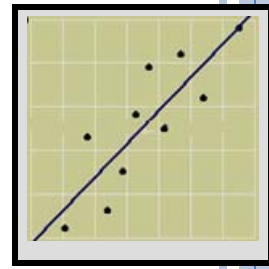


The slide features a dark blue background with a series of vertical stripes in various shades of blue and white on the left side. Several blue circles of different sizes are arranged vertically, some overlapping the stripes and others the main background.

12.5

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Estimation and prediction using the fitted line

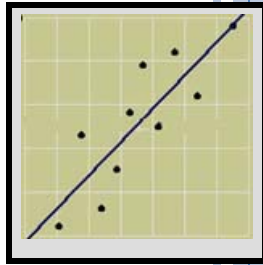


ESTIMATION AND PREDICTION

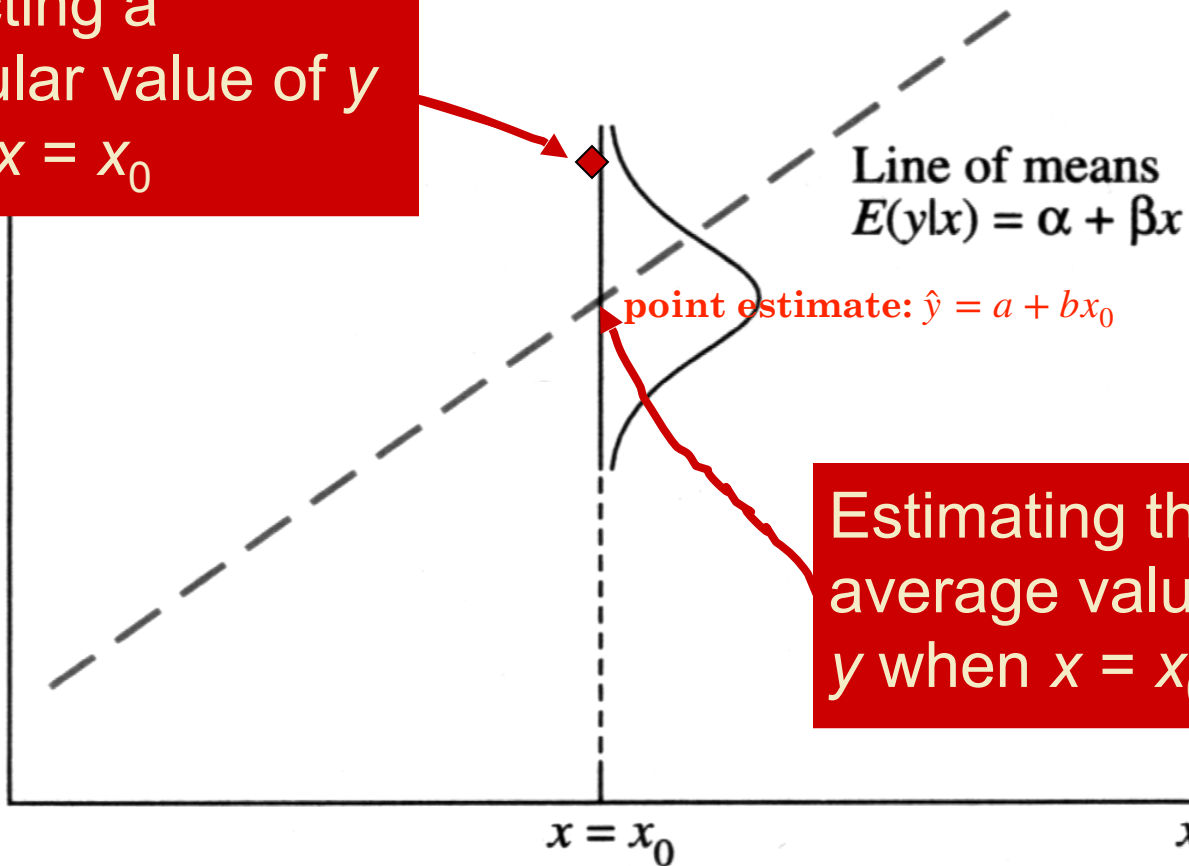
- Once you have
 - ✓ (12.4) used the diagnostic plots to check for violation of the regression assumptions.
 - ✓ (12.2) determined that the regression line is useful ($\beta \neq 0$)
- You are ready to use the regression line to

- ✓ **Estimate** the average value of y for a given value of x : $E(y | x)$
- ✓ **Predict** a particular value of y for a given value of x : y_{new}

ESTIMATION AND PREDICTION



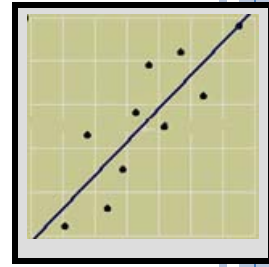
Predicting a particular value of y when $x = x_0$



point estimate: $\hat{y} = a + bx_0$

Estimating the average value of y when $x = x_0$

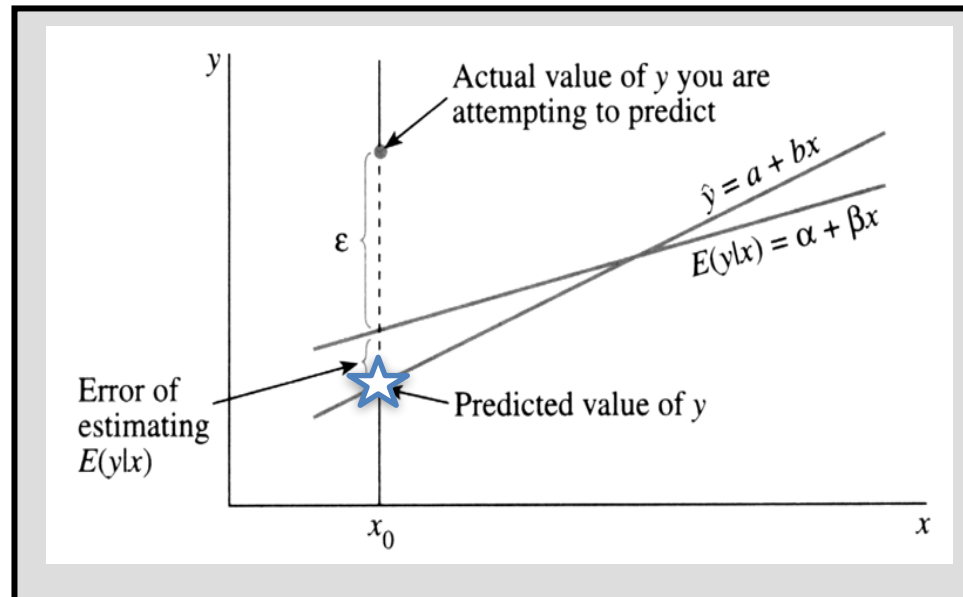
ESTIMATION AND PREDICTION



- The best estimate of either $E(y)$ or y for a given value $x = x_0$ is

$$\hat{y} = a + bx_0$$

- Particular values of y are more difficult to predict, requiring a wider range of values in the prediction interval.



HOW DO WE OBTAIN CI

- Target: $E(y | x_0) = \alpha + \beta x_0$
- We use $\hat{y} = a + bx_0$ to estimate $E(y | x_0)$
- Fact (proof skipped)

$$(a + bx_0) \sim N\left(\alpha + \beta x_0, \sigma^2 \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right)\right)$$

$$\text{Thus, } \frac{(a + bx_0) - (\alpha + \beta x_0)}{\sqrt{MSE\left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right)}} \sim t_{(n-2)}$$

HOW DO WE OBTAIN CI

- To obtain the CI, we start with

$$\begin{aligned} 1 - \alpha &= P \left(-t_{(n-2); \alpha/2} < \frac{(a + bx_0) - (\alpha + \beta x_0)}{\sqrt{MSE(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}})}} < t_{(n-2); \alpha/2} \right) \\ &= P \left(-t_{(n-2); \alpha/2} \sqrt{MSE(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}})} < (a + bx_0) - (\alpha + \beta x_0) < t_{(n-2); \alpha/2} \sqrt{MSE(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}})} \right) \\ &= P \left((a + bx_0) - t_{(n-2); \alpha/2} \sqrt{MSE(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}})} < (\alpha + \beta x_0) < (a + bx_0) + t_{(n-2); \alpha/2} \sqrt{MSE(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}})} \right) \end{aligned}$$

Thus, the $100(1 - \alpha)\%$ CI is

$$(a + bx_0) \pm t_{(n-2); \alpha/2} \sqrt{MSE \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)}$$

HOW DO WE OBTAIN PREDICTION

INTERVAL? ✓ *Predict a particular value of y for a given value of x_0 : y_{new}*

- Target: $y_{new} = \alpha + \beta x_0 + \varepsilon_{new}$
- We use $\hat{y}_{new} = a + bx_0 + 0 = a + bx_0$ to predict y_{new}
- Fact: the forecast error

$$\begin{aligned}\epsilon &= y_{new} - \hat{y}_{new} \\ &= (\alpha + \beta x_0 + \varepsilon_{new}) - (a + bx_0) \\ &\sim N\left(0, \sigma^2 \left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right)\right)\end{aligned}$$

$$\text{Thus, } \frac{(y_{new}) - (\hat{y}_{new})}{\sqrt{MSE \left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right)}} \sim t_{(n-2)}$$

HOW DO WE OBTAIN PREDICTION

INTERVAL? ✓ *Predict a particular value of y for a given value of x_0 : y_{new}*

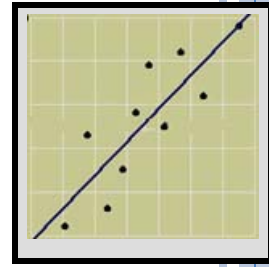
- To obtain the PI, we start with

$$\begin{aligned} 1 - \alpha &= P \left(-t_{(n-2);\alpha/2} < \frac{y_{new} - (\hat{y}_{new})}{\sqrt{MSE(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}})}} < t_{(n-2);\alpha/2} \right) \\ &= P \left(-t_{(n-2);\alpha/2} \sqrt{MSE(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}})} < y_{new} - (\hat{y}_{new}) < t_{(n-2);\alpha/2} \sqrt{MSE(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}})} \right) \\ &= P \left(\hat{y}_{new} - t_{(n-2);\alpha/2} \sqrt{MSE(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}})} < y_{new} < \hat{y}_{new} + t_{(n-2);\alpha/2} \sqrt{MSE(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}})} \right) \end{aligned}$$

The $100(1 - \alpha)\%$ Prediction Interval of y for a given value of x is

$$\hat{y}_{new} \pm t_{(n-2);\alpha/2} \sqrt{MSE(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}})}$$

ESTIMATION AND PREDICTION



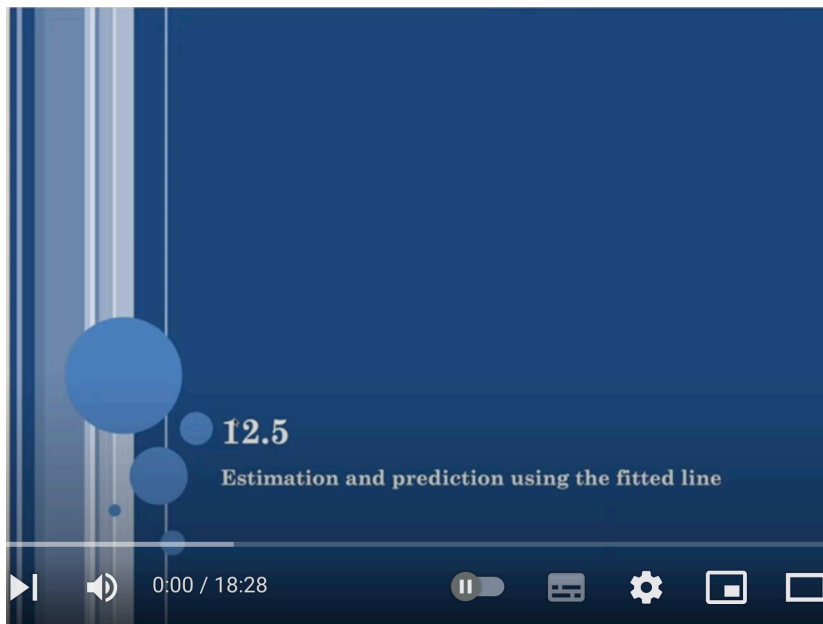
Confidence interval (CI) for the *average value* of y given $x = x_0$

$$(a + bx_0) \pm t_{(n-2), \alpha/2} \sqrt{MSE \left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)}$$

Prediction interval (PI) for a *particular value* of y given $x = x_0$

$$(a + bx_0) \pm t_{(n-2), \alpha/2} \sqrt{MSE \left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right)}$$

$$MSE = \frac{SSE}{n - 2}$$



<https://youtu.be/onFoHVDWhA?si=9X8NB8GdE0sZFUaA>



https://youtu.be/y1LXtX_G1mk?si=I-fs5XB_qR7MwU0i

THE CALCULUS PROBLEM



- Estimate the average calculus grade for students whose achievement score is 50 with a 95% confidence interval.

Calculate $\hat{y} = 40.78424 + .76556(50) = 79.06$

$$\hat{y} \pm 2.306 \sqrt{75.7532 \left(\frac{1}{10} + \frac{(50 - 46)^2}{2474} \right)}$$

79.06 ± 6.55 or 72.51 to 85.61.

THE CALCULUS PROBLEM



- Predict the calculus grade for a **particular student** whose achievement score is 50 with a 95% prediction interval.

$$\text{Calculate } \hat{y} = 40.78424 + .76556(50) = 79.06$$

$$\hat{y} \pm 2.306 \sqrt{75.7532 \left(1 + \frac{1}{10} + \frac{(50 - 46)^2}{2474} \right)}$$

$$79.06 \pm 21.11 \quad \text{or } 57.95 \text{ to } 100.17.$$

Notice how ⁶⁹ much wider this interval is!

MINITAB OUTPUT

Confidence and



Predicted Values for New Observations

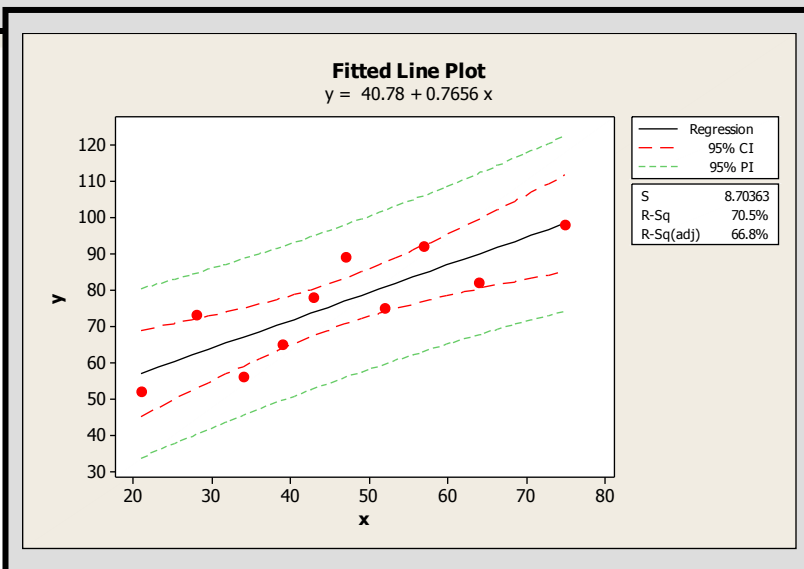
New Obs	Fit	SE Fit	95.0% CI	95.0% PI
1	79.06	2.84	(72.51, 85.61)	(57.95, 100.17)

Values of Predictors for New Observations

New Obs	x
1	50.0

✓ Green prediction bands are always wider than red confidence bands.

✓ Both intervals are narrowest when $x = \bar{x}$.

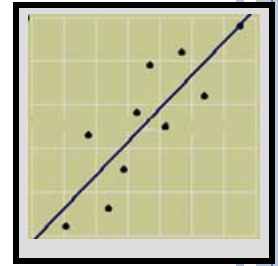


The slide features a dark blue background. On the left side, there are several vertical stripes of varying shades of blue and white. Overlaid on these stripes are several circles of different sizes, also in shades of blue. The largest circle is positioned near the top left, and several smaller circles are arranged vertically below it, some overlapping the stripes.

12.6 CORRELATION ANALYSIS

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CORRELATION ANALYSIS



- The strength of the “linear” relationship between x and y is measured using the **coefficient of correlation**:

$$\text{Correlation coefficient: } r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}$$

- Recall from Chapter 3 that
 - (1) $-1 \leq r \leq 1$
 - (2) r and b have the same sign
 - (3) $r \approx 0$ means no linear relationship
 - (4) $r \approx 1$ or -1 means a strong (+) or (-) **linear** relationship

EXAMPLE



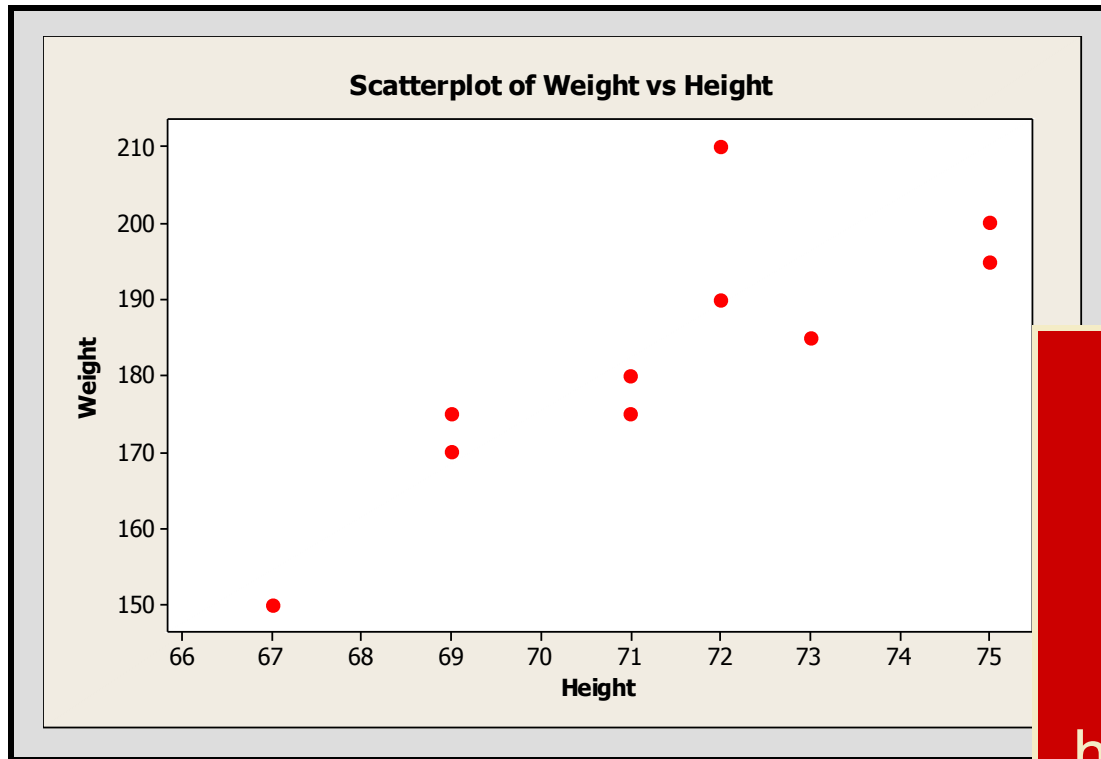
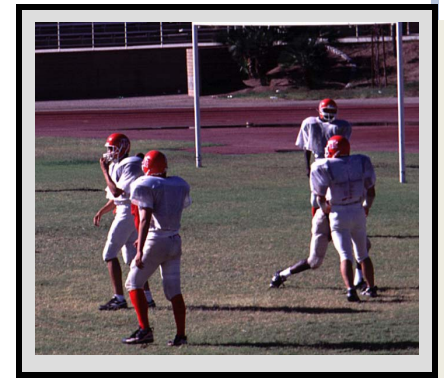
The table shows the heights and weights of $n = 10$ randomly selected college football players.

Player	1	2	3	4	5	6	7	8	9	10
Height, x	73	71	75	72	72	75	67	69	71	69
Weight, y	185	175	200	210	190	195	150	170	180	175

Use your calculator to find the sums and sums of squares.

$$S_{xy} = 328 \quad S_{xx} = 60.4 \quad S_{yy} = 2610$$
$$r = \frac{328}{\sqrt{(60.4)(2610)}} = .8261$$

FOOTBALL PLAYERS



$$r = .8261$$

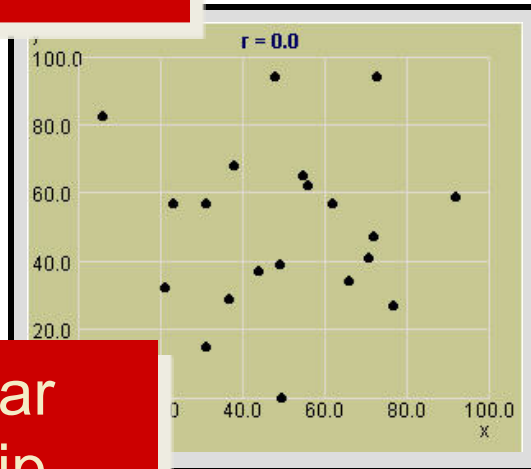
Strong positive
correlation

As the player's
height increases,
so does his
weight.

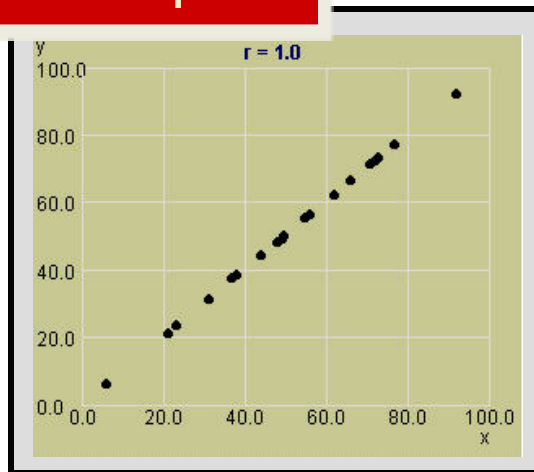
SOME CORRELATION PATTERNS

- Use the **Exploring Correlation** applet to explore some correlation patterns:

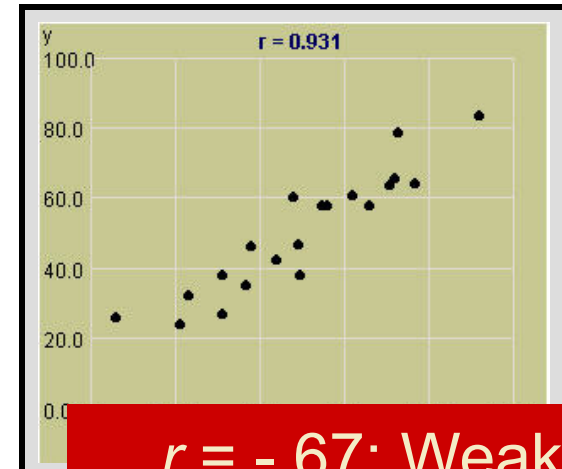
$r = 0$; No (linear) correlation



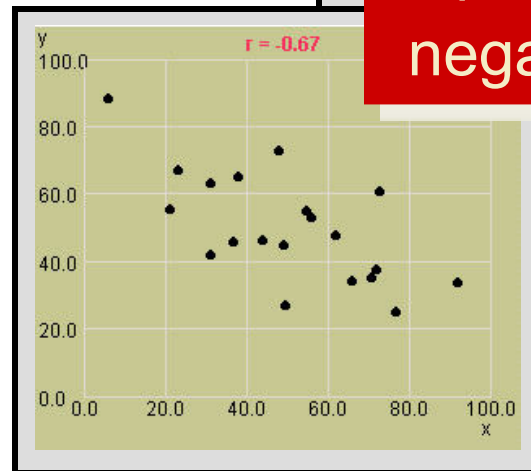
$r = 1$; Linear relationship



$r = .931$; Strong positive correlation



$r = -.67$; Weaker negative correlation



ρ V.S. r

- Population quantity:

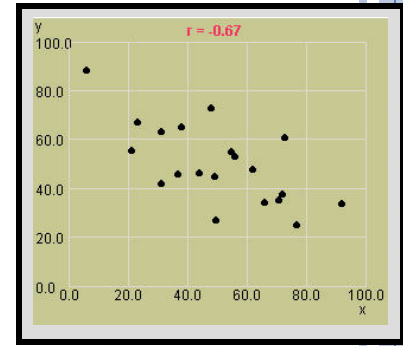
- $$\rho = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} = \frac{E((X - \mu_X)(Y - \mu_Y))}{\sqrt{E(X - \mu_X)^2 E(Y - \mu_Y)^2}}$$

- Sample estimate for ρ

- $$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}.$$

- $$t_{STAT} = r \sqrt{\frac{n-2}{1-r^2}} \sim t_{(n-2)}$$

INFERENCE USING R



- The population **coefficient of correlation** is called ρ (“rho”). We can test for a significant correlation between x and y

To test $H_0 : \rho = 0$ versus $H_a : \rho \neq 0$

Test Statistic:
$$t = r \sqrt{\frac{n-2}{1-r^2}}$$

Reject H_0 if $t > t_{\alpha/2}$ or $t < -t_{\alpha/2}$ with $n-2$ df .

This test is exactly equivalent to the t -test for the slope $\beta=0$.

EQUAL TO THE T-TEST FOR $H_0 : \beta = 0$?

$$\begin{aligned}\left(r\sqrt{\frac{n-2}{1-r^2}}\right)^2 &= r^2 \frac{n-2}{1-r^2} = R^2 \frac{n-2}{1-R^2} \\ &= \frac{(n-2)\text{SSR}/\text{SSTotal}}{\text{SSE}/\text{SSTotal}} \\ &= \frac{\text{SSR}}{\text{SSE}/(n-2)} = \frac{\text{SSR}/1}{\text{SSE}/(n-2)} \\ &= \frac{\text{MSR}}{\text{MSE}} = F_{STAT} = (t_{STAT})^2.\end{aligned}$$

- Recall
 - $r^2 = R^2$
 - $\text{SSE} + \text{SSR} = \text{SSTotal}$
 - $F_{STAT} = (t_{STAT})^2$

$$r = .8261$$

EXAMPLE



Is there a significant positive correlation between weight and height in the population of all college football players?

$$H_0 : \rho = 0$$

$$H_a : \rho > 0$$

$$\begin{aligned} \text{Test Statistic: } t &= r \sqrt{\frac{n-2}{1-r^2}} \\ &= .8261 \sqrt{\frac{8}{1-.8261^2}} = 4.15 \end{aligned}$$

Use the t -table with $n-2 = 8$ df to bound the p -value as p -value $< .005$. There is a significant positive correlation.

KEY CONCEPTS

I. A Linear Probabilistic Model

1. When the data exhibit a linear relationship, the appropriate model is $y = \alpha + \beta x + \varepsilon$.
2. The random error ε has a normal distribution with mean 0 and variance σ^2 .

II. Method of Least Squares

1. Estimates a and b , for α and β , are chosen to minimize SSE, the sum of the squared deviations about the regression line, $\hat{y} = a + bx$.
2. The least squares estimates are $b = S_{xy}/S_{xx}$ and $a = \bar{y} - b\bar{x}$.

KEY CONCEPTS

III. Analysis of Variance

1. Total SS = SSR + SSE, where Total SS = S_{yy} and $SSR = (S_{xy})^2 / S_{xx}$.
2. The best estimate of σ^2 is $MSE = SSE / (n - 2)$.

IV. Testing, Estimation, and Prediction

1. A test for the significance of the linear regression— $H_0 : \beta = 0$ can be implemented using one of two test statistics:

$$t = \frac{b}{\sqrt{MSE / S_{xx}}} \quad \text{or} \quad F = \frac{MSR}{MSE}$$

KEY CONCEPTS

2. The strength of the relationship between x and y can be measured using

$$R^2 = \frac{\text{SSR}}{\text{Total SS}}$$

which gets closer to 1 as the relationship gets stronger.

3. Use **residual plots** to check for nonnormality, inequality of variances, and an incorrectly fit model.
4. **Confidence intervals** can be constructed to estimate the intercept α and slope β of the regression line and to estimate the average value of y , $E(y)$, for a given value of x .
5. **Prediction intervals** can be constructed to predict a particular observation, y , for a given value of x . For a given x , prediction intervals are always wider than confidence intervals.

KEY CONCEPTS

V. Correlation Analysis

1. Use the correlation coefficient to measure the relationship between x and y when both variables are random:

$$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}$$

2. The sign of r indicates the direction of the relationship; r near 0 indicates no linear relationship, and r near 1 or -1 indicates a strong linear relationship.
3. A test of the significance of the correlation coefficient is identical to the test of the slope β .